

Merrill Dmonte

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🌐 [Portfolio](#) | [merrill-dmonte](#) | [Merrill04](#)

SUMMARY

Backend Developer with hands-on experience building full-stack systems using **Node.js, TypeScript, and PostgreSQL**. Skilled in REST API design, JWT authentication, Redis caching, and cloud deployment on AWS with Docker. Passionate about writing clean, scalable code and solving complex problems through strong DSA foundations.

PROJECTS

E-Waste Recycling Management System *Next.js · React.js · MongoDB · Scikit-learn* [GitHub](#) Apr 2025

- Developed a full-stack e-waste recycling platform using **Next.js App Router**, MongoDB, JWT Authentication, and REST APIs, enabling secure user registration, booking management, and facility discovery.
- Integrated **Mapbox GL JS**, Geolocation API, Mapbox Directions API to provide nearest recycling facility search, interactive maps, and optimized navigation routes.
- Built an **AI-powered price prediction system** using Python, Scikit-learn Random Forest, Next.js API Routes, and Google Gemini API to estimate recycling value and provide an intelligent e-waste assistance chatbot.

URL Shortening Service *Node.js · Express.js · PostgreSQL · Redis · React.js · JWT* [GitHub](#) July 2026

- Built a full-stack URL shortener with **Node.js/Express REST API** and React frontend, implementing JWT authentication with access/refresh token rotation and bcrypt password hashing for secure multi-user session management.
- Engineered a **Redis caching layer** on the redirect endpoint reducing database load, with PostgreSQL as persistent store using indexed short-code lookups, db-migrations runner, and cascaded relational schema across users, URLs, and click events.
- Developed a **click analytics pipeline** capturing device type, browser, and referrer per redirect via async UA parsing, aggregated using PostgreSQL window functions and served as daily/weekly/monthly timeseries with Recharts visualizations on the frontend.

EDUCATION

Fr. Conceicao Rodrigues College of Engineering, Mumbai University 2022 – 2026

BE Computer Engineering — CGPA: 8.01

- Coursework:** Data Structures, Computer Networks, Operating Systems, DBMS, Machine Learning

SKILLS

Languages : JavaScript, TypeScript, Java, C++, Python

Backend : Node.js, Express.js, REST APIs, JWT Authentication

Frontend : React.js, Next.js, Tailwind CSS

Databases : PostgreSQL, MongoDB, Redis

Cloud & DevOps : AWS (EC2, S3, Lambda), Docker, Git, Postman

Concepts : Data Structures & Algorithms, System Design, OOP, Microservices, CI/CD

CERTIFICATIONS & ACHIEVEMENTS

AWS Academy Graduate — [Cloud Architecting - Training Badge](#)

AWS Academy Graduate — [Machine Learning for Natural Language Processing - Training Badge](#)

AWS Academy Graduate — [Microservices and CI/CD Pipeline Builder - Training Badge](#)

AWS Academy Graduate — [Cloud Web Application Builder - Training Badge](#)

Solved 200+ DSA problems on [LeetCode](#) and [Codeforces](#)